Worksheet 10.1

Science Quest 9: pages 344–8

Conduction and convection

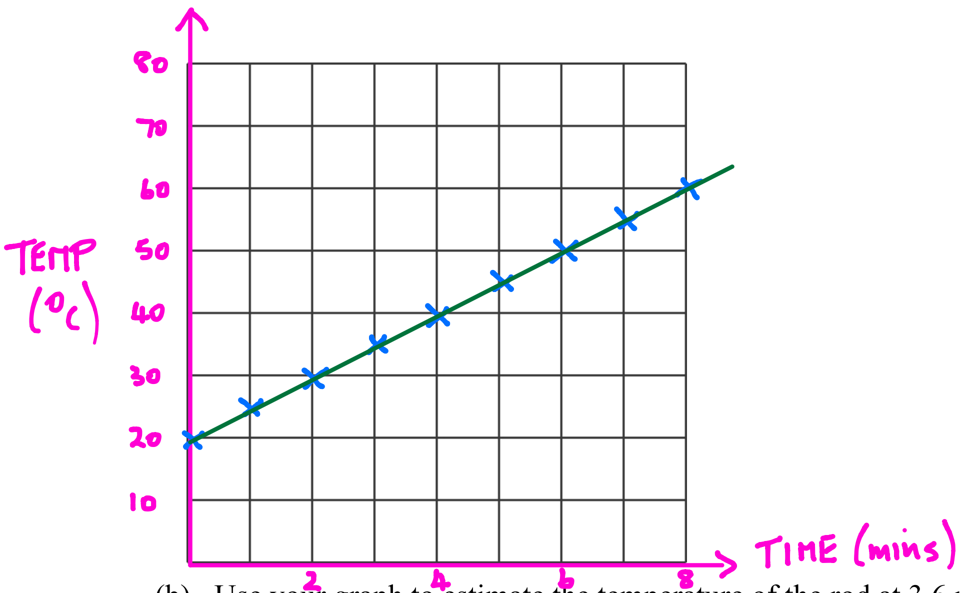
Student: .... Class: ....

Testing conduction

1. The heat conduction properties of a new material were tested. One end of a 100-cm length of the material was heated, and the temperature of the rod at the other end was recorded every minute. The data is tabulated below.

(a) Plot a line graph of the data and draw the line of best fit. (Start your graph at 20 °C).

Time (min)	Temperature (°C)
0	20.0
1	25.0
2	29.8
3	35.2
4	40.0
5	44.7
6	50.3
7	54.9
8	59.9



(b) Use your graph to estimate the temperature of the rod at 3.6 minutes after the start.

38 °C

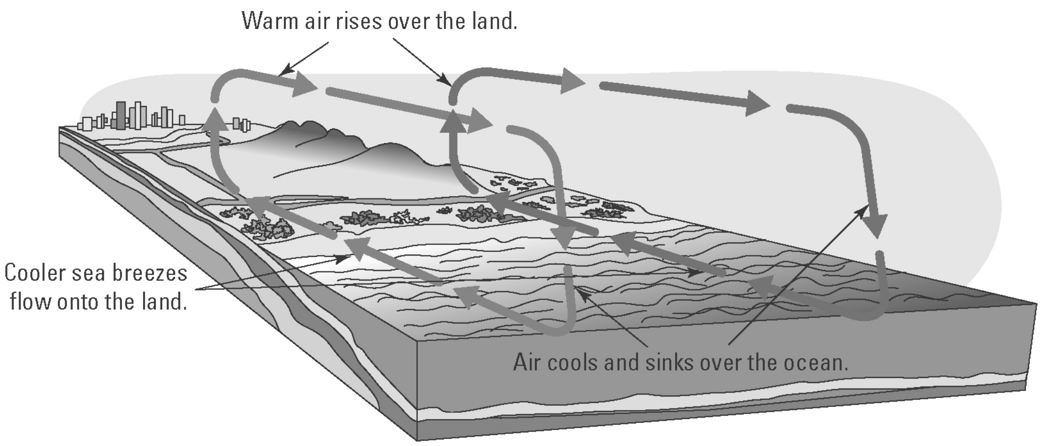
(c) Identify the dependent and independent variables in this experiment.

Dependent – temperature

Independent – time

Convection

2. Consider the following illustration showing the movement of air near the ocean shore. Describe how the production of convection currents makes the climate near the coast milder than inland.



- Warm air rises above land and moves out to sea.
- Cool air from the sea moves in over the land to take the place of the rising warm air.
- The "sea breeze" cools everything down.